

DD_ASR-02 - Information Request

Version: 1.0 Status: Draft for review Bundle: 10_other Audience: customer care developers, ticketing developers, frontend developers, QA Related DDs:

- DD_TCK-01 - Ticketing and Case Management
- DD_CUST-INT-01 - Customer Interaction Timeline
- DD_ILM-CFG-01 - Customer Service
- DD_RLM-CFG-01 - HomePass Configuration
- DD_SIP-01 - Package Management
- DD_PLM-CFG-01 - Service Catalog
- DD_BIL-02-READ-01 - Invoice Read API
- FE-APP-01 - BSS Backoffice Web App
- FE-APP-04 - Customer Self Care Frontoffice App

MVP simplification baseline: ASR-02 is implemented as TCK-01 information-request ticket categories, templates, queues, and closure rules. It is not a separate ticketing service, database, or deployment in the first release.

1. Purpose

ASR-02 defines the information request flow for customer questions that require a trackable answer but do not start as a technical fault, complaint, or operational service request.

Examples:

- customer asks which package they are subscribed to;
- customer asks whether an address is covered;
- customer asks why a package is not available at their HomePass;
- customer asks for an explanation of invoice lines;
- customer asks about payment channels;
- customer asks about KYC status;
- customer asks about installation/order status;
- customer asks for a document or reference already available in SOPHIX.

This DD does not create a new ticketing engine. Information requests are implemented as configured TCK ticket types/categories, with ASR-specific answer templates, research tasks, and closure rules.

2. Ownership Boundary

Concern	Owner
Ticket lifecycle, comments, assignment, SLA, attachments	TCK-01
Information request categories, research checklist, answer policy	ASR-02 config inside TCK/ASR
Customer/account data	ILM-CFG-01
Coverage/HomePass information	RLM-CFG-01
Package/service explanations	SIP-01 / PLM-CFG-01
Invoice/payment read-only explanation	BIL read APIs
Actual billing correction or adjustment	BIL-02-ADJ-01, not ASR-02
Technical repair	ASR-01 / WO
Complaint handling	ASR-03
Customer timeline projection	CUST-INT-01

Rule: ASR-02 is read-only with respect to business records. It may create a ticket, comments, and timeline entries, but it must not change subscription, billing, package, HomePass, or KYC records.

3. When To Use This Flow

Use ASR-02 when the expected outcome is an answer, explanation, document/reference, or routing guidance.

Do not use ASR-02 when:

Scenario	Use
Customer reports service not working	ASR-01 Technical Trouble
Customer is unhappy and requests correction/escalation	ASR-03 Complaint
Customer requests an action such as address change, SIP reset, equipment swap, or package change	ASR-04 Service Request or owning workflow
Customer disputes an invoice and wants a correction	TCK billing complaint plus BIL adjustment flow

4. High Level Flow

```
flowchart TD
    A["Customer asks question"]
    B["Create TCK ticket<br/>GENERAL_INQUIRY or PACKAGE_OR_CATALOG_QUERY"]
    C["Resolve category guidance"]
    D{"Can answer immediately?"}
    E["Add customer-visible answer"]
    F["Resolve ticket"]
    G["Assign research task/queue"]
    H["Gather data from owning APIs"]
    I{"Needs different process?"}
    J["Convert/link to ASR-01/03/04 or BIL flow"]
    K["Prepare answer"]

    A --> B --> C --> D
    D -->|Yes| E --> F
    D -->|No| G --> H --> I
    I -->|Yes| J
    I -->|No| K --> E --> F
```

5. Ticket Types And Categories

ASR-02 primarily uses these TCK ticket types:

TCK type	Use
GENERAL_INQUIRY	Generic question not tied to a specific catalog/billing/KYC domain.
PACKAGE_OR_CATALOG_QUERY	Package, product, speed, eligibility, price, or coverage explanation.
KYC_SUPPORT	KYC status/document explanation, if no correction is requested.
BILLING_COMPLAINT	Use only when customer asks billing explanation. If correction is requested, route to complaint/adjustment.

Recommended categories:

Category code	Meaning	Typical answer source	Default priority
PACKAGE_EXPLANATION	Explain current package or available packages.	SIP/PLM	P4
COVERAGE_INFORMATION	Explain coverage or HomePass availability.	RLM	P4
ORDER_STATUS_INFORMATION	Explain current order/install status.	FUL/WO	P4
INVOICE_EXPLANATION	Explain invoice lines or taxes.	BIL read APIs	P3
PAYMENT_CHANNEL_INFORMATION	Explain payment channels.	PAY-GW/BIL-05	P4
KYC_STATUS_INFORMATION	Explain KYC state or missing documents.	ILM/FUL KYC	P4

Category code	Meaning	Typical answer source	Default priority
ACCOUNT_REFERENCE_INFORMATION	Provide account/customer references.	ILM	P4
GENERAL_QUESTION	Other informational question.	Agent/research	P4

WO creation is not allowed by default for ASR-02 categories.

6. ASR-Specific Tables

6.1 asr_info_category_config

Stores behavior for each information request category.

Use this table to decide answer source, default queue, whether research is required, and whether self-care can create this request.

Column	Type	Purpose	Sample
config_id	UUID PK	Config id.	airc-001
operator_code	TEXT	Operator/tenant.	WIK
category_code	TEXT	TCK category.	PACKAGE_EXPLANATION
default_type_code	TEXT	TCK type.	PACKAGE_OR_CATALOG_QUERY
answer_source_module	TEXT	Primary source module.	SIP-01
research_required	BOOLEAN	Whether ticket needs research before answer.	false
default_queue_code	TEXT	Queue if not immediate.	CARE_L1
selfcare_allowed	BOOLEAN	Customer can create from self-care.	true
answer_template_code	TEXT NULL	Suggested template.	PACKAGE_EXPLANATION_BASIC
conversion_policy_json	JSONB	When to convert/escalate.	{"billingCorrection":"ASR-03"}
status	TEXT	ACTIVE, RETIRED.	ACTIVE

Recommended indexes:

- Unique (operator_code, category_code).

6.2 asr_info_answer_template

Stores safe **agent answer templates** for common information requests.

This table is not a customer-notification template catalog. It contains canned-answer content shown to support agents inside TCK/ASR screens so they can answer consistently. If an answer must be pushed to a customer through SMS, email, WhatsApp, portal notification, or another delivery channel, ASR passes the selected answer text or event-kind hint to NOT-01. NOT-01 still owns delivery template, channel, language, quiet hours, fallback, and retry policy.

Use this table to make answers consistent and customer-safe.

Column	Type	Purpose	Sample
template_id	UUID PK	Template id.	ait-001
operator_code	TEXT	Operator/tenant.	WIK
template_code	TEXT	Stable template code.	PACKAGE_EXPLANATION_BASIC
language_code	TEXT	Language.	en
subject_template	TEXT	Suggested subject.	Your current package information
body_template	TEXT	Answer body with variables.	Your current package is {{packageName}}.
required_variables_json	JSONB	Variables required.	["packageName", "monthlyPrice"]
status	TEXT	ACTIVE, RETIRED.	ACTIVE

Recommended indexes:

- Unique (operator_code, template_code, language_code).

6.3 asr_info_research_task

Stores research work attached to a ticket.

Use this table when the agent cannot answer immediately and must wait for another team or API result.

Column	Type	Purpose	Sample
research_task_id	UUID PK	Research task id.	airt-001
ticket_id	TEXT	TCK ticket id.	tck-001
operator_code	TEXT	Operator/tenant.	WIK
research_type	TEXT	API_LOOKUP, TEAM_REVIEW, DOCUMENT_RETRIEVAL.	API_LOOKUP
source_module	TEXT	Module/team to consult.	RLM-CFG-01
status	TEXT	OPEN, IN_PROGRESS, COMPLETED, FAILED, CANCELLED.	OPEN
request_payload_json	JSONB	Research request details.	{"homepassId": "HP-300045"}
result_payload_json	JSONB NULL	Research result.	{"sellable": true}
assigned_queue_code	TEXT NULL	Queue if manual review.	CARE_L2
created_at	TIMESTAMP	Creation timestamp.	2026-06-03T10:00:00Z
completed_at	TIMESTAMP NULL	Completion timestamp.	NULL

Recommended indexes:

- (ticket_id, status).
- (operator_code, assigned_queue_code, status).

6.4 asr_info_answer

Stores the final structured answer given to the customer.

Use this table to preserve what was answered and which source data supported it.

Column	Type	Purpose	Sample
answer_id	UUID PK	Answer id.	aia-001
ticket_id	TEXT	TCK ticket id.	tck-001
operator_code	TEXT	Operator/tenant.	WIK
template_code	TEXT NULL	Template used.	PACKAGE_EXPLANATION_BASIC
language_code	TEXT	Language.	en
answer_body	TEXT	Customer-visible answer.	Your current package is Fiber 100M.
source_refs_json	JSONB	Modules/entities used.	[{"module": "SIP-01", "ref": "pkg-100m"}]
answered_by_user_id	TEXT	Agent/system.	usr-care-001
answered_at	TIMESTAMP	Answer timestamp.	2026-06-03T10:08:00Z

Recommended indexes:

- (ticket_id, answered_at).

7. APIs

7.1 Create Information Request Ticket

Uses TCK create ticket API.

```
POST /api/tickets
Idempotency-Key: selfcare-info-20260603-001
```

Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "typeCode": "PACKAGE_OR_CATALOG_QUERY",
  "categoryCode": "PACKAGE_EXPLANATION",
  "sourceChannel": "SELF_CARE",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "subject": "Package information request",
  "description": "Customer wants to know current package and price.",
  "customerVisible": true
}
```

7.2 Get Guidance

GET /api/asr/information-requests/{ticket_id}/guidance

Response:

```
{
  "ticketId": "tck-20260603-000020",
  "categoryCode": "PACKAGE_EXPLANATION",
  "answerSourceModule": "SIP-01",
  "researchRequired": false,
  "answerTemplateCode": "PACKAGE_EXPLANATION_BASIC",
  "allowedActions": ["ANSWER_AND_RESOLVE", "CREATE_RESEARCH_TASK", "CONVERT_TO_COMPLAINT"]
}
```

7.3 Create Research Task

POST /api/asr/information-requests/{ticket_id}/research-tasks
Content-Type: application/json

```
{
  "researchType": "API_LOOKUP",
  "sourceModule": "RLM-CFG-01",
  "requestPayload": {
    "homepassId": "HP-300045",
    "question": "Why is package Fiber 1G not available?"
  }
}
```

7.4 Answer And Resolve

POST /api/asr/information-requests/{ticket_id}/answer-and-resolve
Content-Type: application/json

```
{
  "templateCode": "PACKAGE_EXPLANATION_BASIC",
  "languageCode": "en",
  "answerBody": "Your current package is Fiber 100M. The monthly recurring charge is 4,500 KES before any active discounts",
  "sourceRefs": [
    { "module": "SIP-01", "ref": "pkg-fiber-100m" },
    { "module": "SUB-LM", "ref": "SUB-700045" }
  ],
  "resolutionCode": "ANSWER_PROVIDED"
}
```

This API:

1. Stores asr_info_answer.
2. Adds customer-visible TCK comment.
3. Resolves the TCK ticket.
4. Emits InformationRequestAnswered.

7.5 Convert To Another Flow

POST /api/asr/information-requests/{ticket_id}/convert
Content-Type: application/json

```
{
  "targetFlow": "ASR-03",
  "targetCategoryCode": "BILLING_DISPUTE",
  "reason": "Customer does not accept invoice explanation and requests correction."
}
```

Conversion should not delete the original ticket history. Preferred implementation:

- update current ticket category/type if TCK policy allows; or
- create a linked new ticket and close original as CONVERTED.

8. Events

Event	Topic	When emitted
InformationRequestCreated	sophix.asr.info.created	Information request ticket created.
InformationRequestResearchStarted	sophix.asr.info.researchstarted	Research task created.
InformationRequestResearchCompleted	sophix.asr.info.researchcompleted	Research task completed.
InformationRequestAnswered	sophix.asr.info.answered	Customer-visible answer provided.
InformationRequestConverted	sophix.asr.info.converted	Request converted/linked to another flow.

TCK still emits generic ticket events. ASR-02 events are business-specific for reporting and customer timeline.

9. Business Rules

Rule ID	Rule	Notes
R-ASR-INF0-01	Information request must use a configured TCK category.	Avoid free-form routing.
R-ASR-INF0-02	ASR-02 is read-only against business modules.	No subscription/billing mutations.
R-ASR-INF0-03	Customer-visible answers must not include internal diagnostics, staff notes, or raw system errors.	Safety.
R-ASR-INF0-04	If the customer requests correction/compensation, convert or link to complaint/adjustment flow.	Avoid hiding complaints as questions.
R-ASR-INF0-05	Research tasks must name the source module/team and preserve result payload.	Audit.
R-ASR-INF0-06	Self-care can create only categories with <code>selfcare_allowed = true</code> .	Prevent unsupported public flows.
R-ASR-INF0-07	Answer-and-resolve requires a customer-visible answer body.	Closure must be explainable.
R-ASR-INF0-08	Ticket resolution must use TCK status APIs.	TCK remains source of truth.

10. Common Source APIs

ASR-02 does not own the data it explains. It calls owning module APIs.

Question	Source module/API
Current package	SUB-LM subscription detail, SIP-01 package detail.
Available packages	SIP-01 package APIs, RLM coverage/sellability APIs.
Coverage/HomePass	RLM-CFG-01 HomePass search/detail.
Invoice line explanation	BIL-02-READ-01 invoice detail/read APIs.
Payment channel explanation	PAY-GW-01 channel config, BIL-05 wallet topup channels.
KYC status	ILM/FUL KYC status APIs.
Order/install status	FUL-02 order APIs, WO status APIs where linked.

11. Frontend Usage

11.1 FE-APP-01 Backoffice

Backoffice uses ASR-02 for:

- Create information request from customer 360.
- Show category-specific answer guidance.
- Launch module lookups from the ticket.
- Create research task.
- Draft customer-visible answer.
- Answer and resolve.
- Convert to complaint, technical trouble, service request, or billing adjustment when needed.

11.2 FE-APP-04 Customer Self-Care

Self-care may expose:

- Ask a question;
- select topic;
- attach optional screenshot/document;
- view answer and ticket status;
- add customer-visible follow-up comment.

Self-care must only display customer-visible comments and final answers.

12. Infra Recommendation

ASR-02 should run inside the TCK service/module. No separate pod is required for a medium budget deployment.

Component	Recommendation
API	Existing TCK API pods.
DB	Existing TCK schema with ASR info tables.
Workers	Existing TCK worker pool.
Cache	Optional cache for category/template config.

AWS deployment follows TCK deployment: same ECS/EKS service, RDS schema, Kafka/MSK, and CloudWatch monitoring.

Alarms:

- research task backlog above threshold;
- answer SLA breach count;
- repeated conversion from information request to complaint, indicating poor first-contact answer quality.

13. Testing Requirements

13.1 Unit Tests

- Category guidance resolves source module and template.
- Self-care blocked for category where `selfcare_allowed = false`.
- Answer-and-resolve requires answer body.
- Convert action preserves original ticket history.
- Customer-visible answer redaction validation blocks internal details.

13.2 Integration Tests

- Create package information ticket.
- Fetch guidance.

- Answer and resolve with SIP/SUB source refs.
- Create research task and complete it.
- Convert invoice explanation to complaint flow.

13.3 End To End Tests

- Customer asks package question in self-care, CSR answers, customer sees answer.
- CSR cannot close information request without customer-visible answer.
- Customer asks invoice explanation, then disputes invoice, ticket converts to complaint/adjustment path.

14. Definition Of Done

ASR-02 is complete when:

- Information request categories are seeded in TCK.
- ASR info category config exists.
- Answer templates exist for common questions.
- Guidance API works.
- Research task flow works.
- Answer-and-resolve stores structured answer and resolves ticket through TCK.
- Conversion/linking to other ASR flows works.
- Customer timeline receives information request events.
- Backoffice and self-care can execute the intended flows.